Bhendi diseases

1. Powdery mildew - *Erysiphe cichoracearum*
2. Cercospora leaf spot - *Cercospora abelmoschi*, *C. hibiscina* and *C. malayensis*
3. Alternaria leaf spot - *Alternaria hibiscinum*
4. Fusarium wilt - *Fusarium oxysporum* f.sp. *vasinfectum*
5. Yellow Vein Mosaic - *Yellow Vein Mosaic Virus* (YVMV)

Powdery mildew - *Erysiphe cichoracearum*

Powdery mildew - *Erysiphe cichoracearum*

- Powdery mildew is an important disease of this crop and under favourable weather condition causes significant yield reductions particularly if the infection takes place at in early stages of plant growth.

- **Severe in shade and close plantings**

Symptoms:

- Symptoms first appear as **minute discoloured patches** with thin fine mesh work of white mycelium arising at many places on the **upper surface of lower leaves**

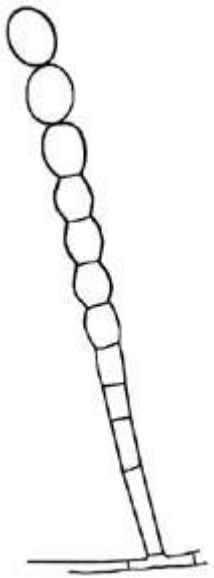
- These white patches soon join together to form larger **white greyish powdery coating** discernible on the severely affected leaves and in later stages the affected leaves turn yellow and finally drop.

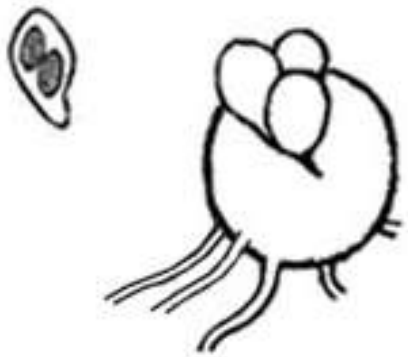
Powdery mildew - *Erysiphe cichoracearum*

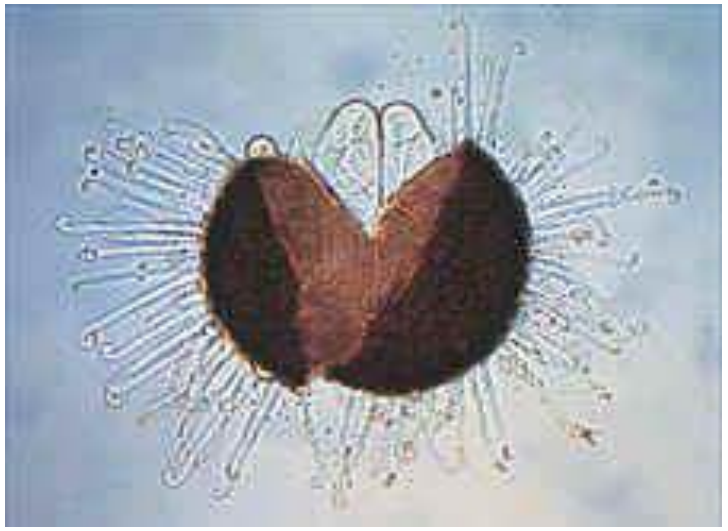
Pathogen:

- The conidia are **single celled, hyaline, barrel shaped and in long chains**.
- The conidial dimensions vary with the physiologic race and the host.
- **Cleistothecia are globose**, dark with hyaline to dark brown and mycelioid appendages.
- They contain **8 to 18 asci and the asci are pedicellate**, ovate to broadly ovate or ellipsoid.
- The number of **ascospores per ascus is usually two, rarely three**. The ascospores are **one celled and hyaline oval to sub-cylindrical**.

Powdery mildew - *Erysiphe cichoracearum*







Powdery mildew - *Erysiphe cichoracearum*

Disease cycle and epidemiology:

- The **primary infection** of the leaves is caused by **wind blown ascospores** (wherever cleistothecia are present) **or by conidia** formed on the earlier sown crop/ indoor cultivation in the neighbouring areas.

- In the process of infection, conidia germinate and after penetration, colonization of the entire leaf takes place.

- The conidia formed abundantly on primary infections are blown by wind and air currents and cause secondary infections and the cycle is repeated.

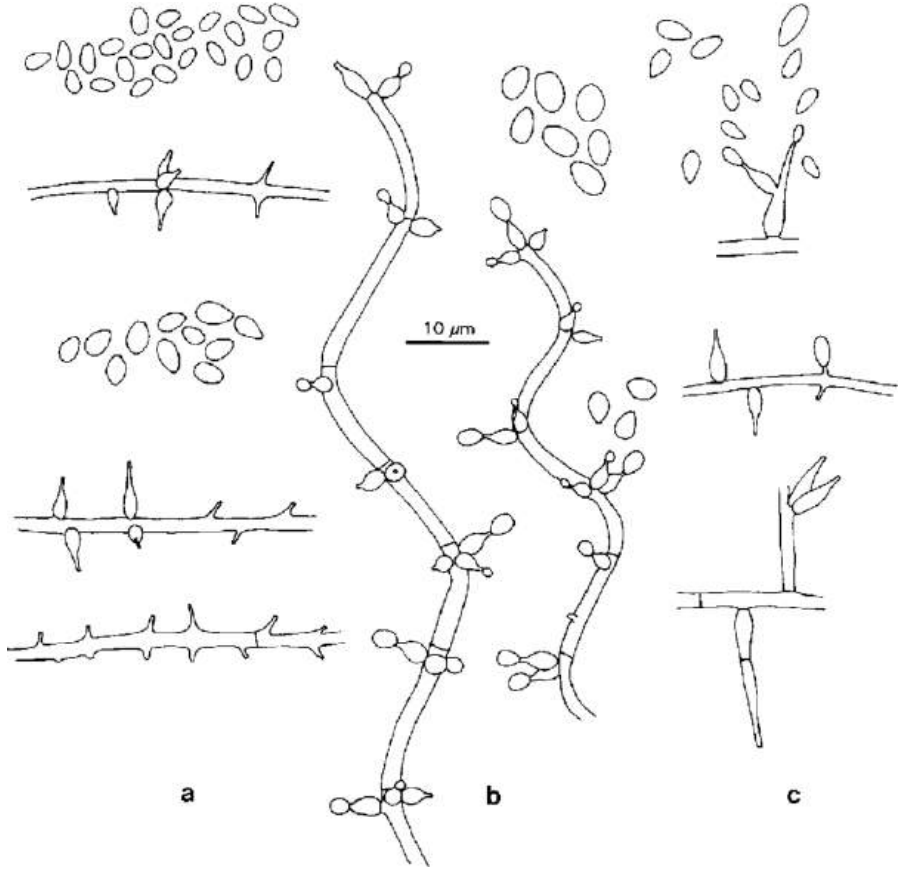
- The pathogen requires 60-80 per cent relative humidity for the development of the disease and **dry conditions for the growth and sporulation of the fungus.**

Powdery mildew - *Erysiphe cichoracearum*

Management

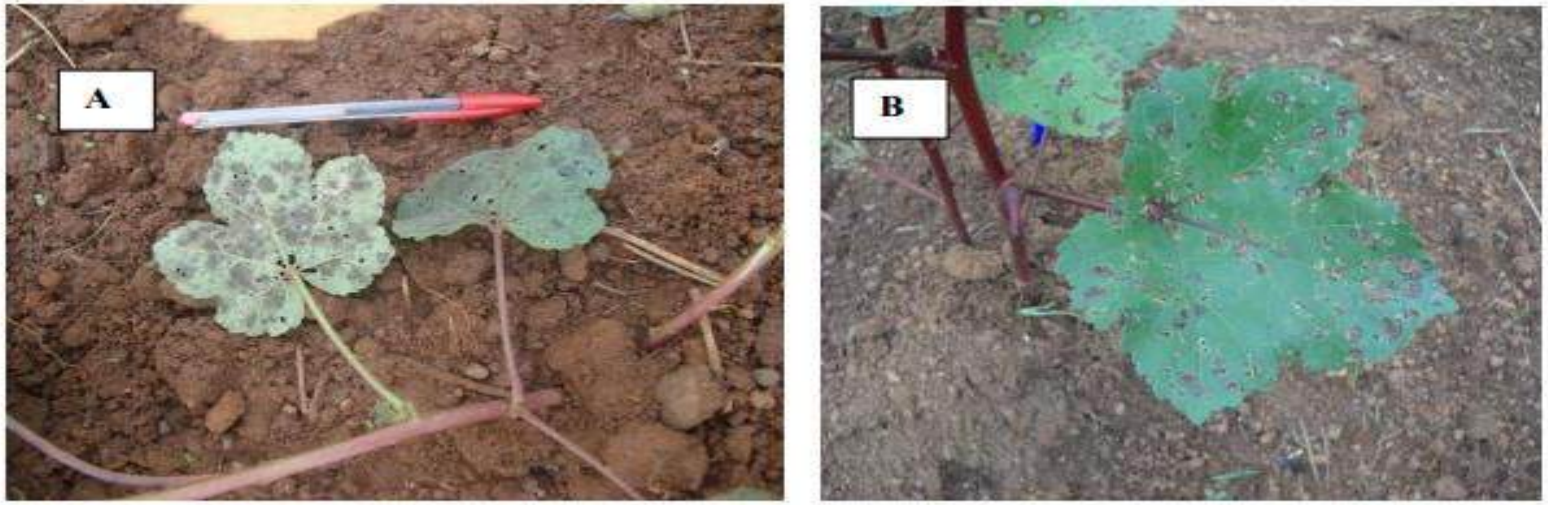
- Moderately rest. Var. , **Early long green, Long green, Red bhendi and Vaishali , Vadhu, AE37, AE 52, AE 79 and BH 102** may be cultivated
- With the initiation of the disease spray the crop with fungicides like wettable sulphur (0.25%) or dinocap (0.05%) or carbendazim (0.05%) or hexaconazole (0.05%) or difenoconazole (0.03%) and repeat at 10 to 14 days interval.
- Biocontrol agent –A hyperparasitic fungus, ***Aphanocladium album*** which develops aerial mycelium and conidia, but it has no sexual stage
- Spraying of neem oil and NSKE extract

Biocontrol agent on powdery mildew
Aphanocladium album



Cercospora leaf spot - *Cercospora abelmoschi, C. hibiscina, C. malayensis*

Cercospora abelmoschi causes **sooty black, angular spots** on lower surface of leaves



C. malayensis **usually starts with small brown irregular spots surrounded by red or pink coloration**

Figure 1 : Symptoms characteristic of the cercosporiose due to *Pseudocercospora abelmoschii* (A) et à *Cercospora malayensis* (B).

ease symptom on leaf: Photo by : SKSain

Cercospora leaf spot - *Cercospora abelmoschi, C. hibiscina, C. malayensis*

The leaf spot cause severe defoliation and are common during humid seasons

Symptoms

- Three different species of *Cercospora* are associated with this disease. The symptoms of each species are described below:

Cercospora abelmoschi:

- The fungus produces angular spots leaf spots but grows as a sooty mould on the lower surface of the leaves
- Severely affected leaves roll, wilt and fall down to the ground.

Cercospora leaf spot - *Cercospora abelmoschi, C. hibiscina, C. malayensis*

Symptoms contd..

***Cercospora malayensis*:**

The spots caused by this species produce irregular leaf spots with grey centers and red to purple borders.

***Cercospora hibiscina*:**

The fungus produces dark olivaceous patches of mould on the lower surface of the leaves.

Cercospora leaf spot - *Cercospora abelmoschi, C. hibiscina, C. malayensis*

Disease cycle and epidemiology:

- The fungus overwinters in **the infected plant debris as conidia or stromata in soil and also on the wild species of *Abelmoschus***.
- The conidia in favourable weather conditions, germinate and cause infection of the host through stomata.
- The spores produced on the primary spots are blown by wind and cause secondary infections.
- The three species cause infection in moderate temperatures (25-29° C) and high humidity.

Cercospora leaf spot - *Cercospora abelmoschi, C. hibiscina, C. malayensis*

Management:

- Host plant resistance - **Long green , Round selection, IC3769,4685 and 6316**
- Cultural practices like collection and destruction of infected plant debris, crop rotation and destruction of wild hosts from in and around the field should be followed to keep the disease under check.
- With the initiation of the disease, spray the crop with fungicides like carbendazim (0.1%) or benomyl (0.1%) or mancozeb (0.25%) or chlorothalonil (0.2%) and repeat at 10 to 14 days interval.

Alternaria leaf spot - *Alternaria hibiscinum*

Alternaria blight of okra pods under field diseased okra pod (right) compare healthy

Fig. 3. Inner symptoms of diseased okra pod of cross section showing different degree of necrotic tissues of diseased pods invaded by *A. alternata* compared with control (left)

Alternaria leaf spot - *Alternaria hibiscinum*

Symptoms

- Spots with **brown in colour with sub circular of varying sizes**.
- Spots contains concentric rings appearing like a target board

Disease cycle & Management

Refer to *Alternaria* in previous crop

Fusarium wilt - *Fusarium oxysporum* f.sp. *vasinfectum*

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The disease appear on any stage of plant growth, crop **planted during summer months suffer more than the crop** planted during February-March.

Symptoms:

- The characteristic symptoms of the disease appear as **yellowing and stunting of the plants followed by wilting and rolling of the leaves** and finally the plant dies.
- Before the appearance of typical wilting symptoms, the leaves hang down during day time and recover again in the night but ultimately they wilt and die.

Fusarium wilt - *Fusarium oxysporum* f.sp. *vasinfectum*

Symptoms contd…

- Vascular bundles of the affected plants appear as dark streaks and the whole stem is blackened in case of severe infection.

Disease cycle and epidemiology:

- The fungus survives from one season to other in the form of chlamydospores and in infected seeds.
- In contact with the host roots, the chlamydospores or conidia germinate and penetrate.
- The pathogen proliferates extensively in the cortical tissues, extending up and down.

Fusarium wilt - *Fusarium oxysporum* f.sp. *vasinfectum*

Disease cycle and epidemiology:

- After some growth in root cortex, the pathogen reaches the xylem vessels where it multiplies very rapidly and result in browning of xylem vessels and clogged further.

- The optimum temperature for growth of the fungus is 25°C and for disease development ranges between 22 – 28° C.

- Moisture is not that much important for infection.

Fusarium wilt - *Fusarium oxysporum* f.sp. *vasinfectum*

Management:

- Cultural practices like long crop rotations, exposing the soil to the sun during summer months by deep ploughing, soil solarization and destruction of diseased roots are some of the practices which can reduce the disease.

- Some cvs. of okra **like Pusa Sawani and Pusa Makhmali, IC 6653,7594** have been reported resistant to this disease.

- ***A. manihot*** (wild relative of okra) can be used in breeding programme

- Use healthy seed and treat with carbendazim (0.2%) or Ziram (0.3%) or **hot water treatment at 45 ±1 ^{o} C /30 min** was effective

Yellow Vein Mosaic - *Yellow Vein Mosaic Virus* (YVMV)

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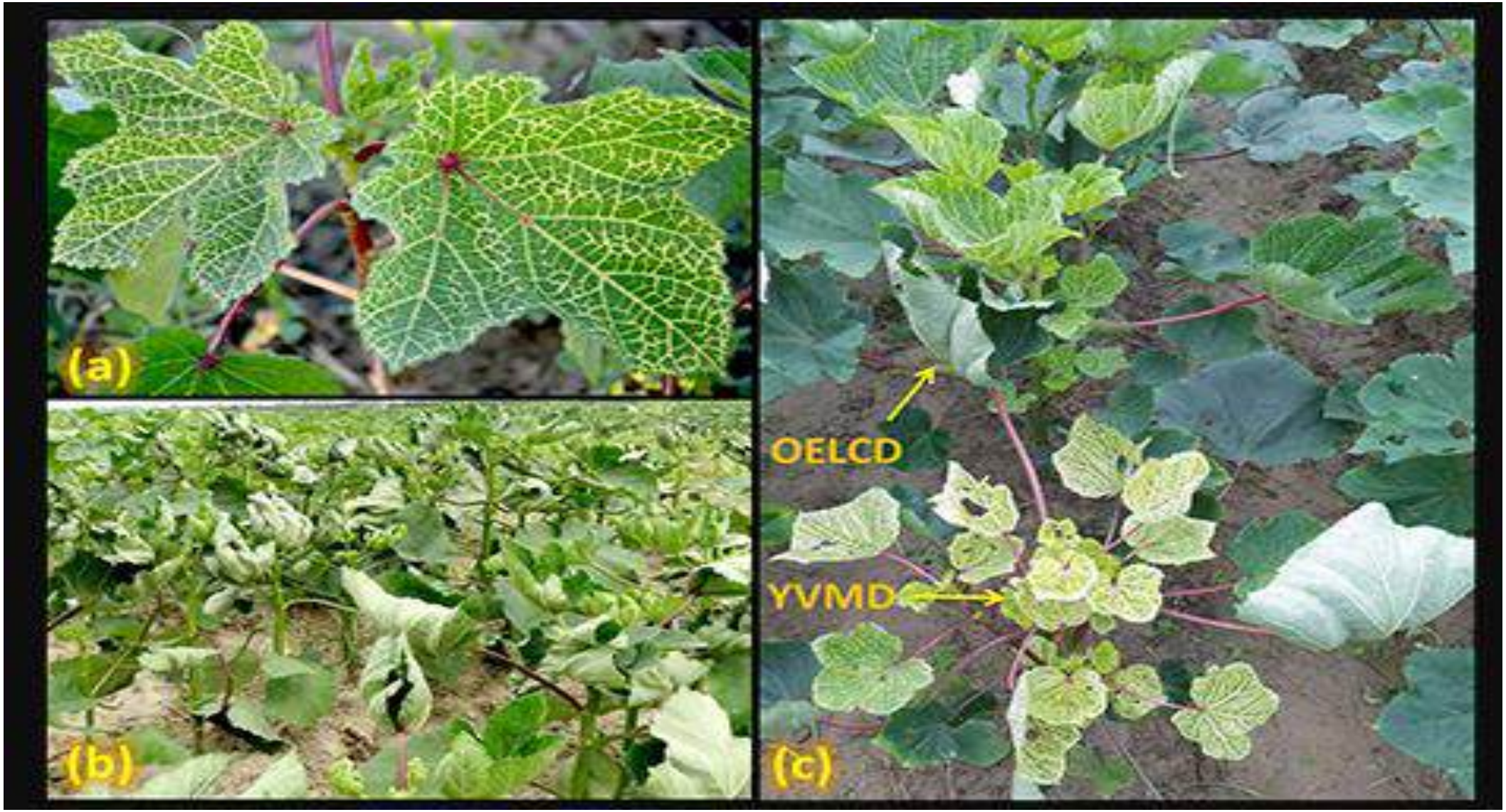

Yellow Vein Mosaic - *Yellow Vein Mosaic Virus* (YVMV)

Symptoms:

- **Very destructive disease** if infection takes place in the early stage of the crop growth
- Yield losses may range from **49.3 to 93.8%** depending the stage of infection
- The characteristic symptoms include vein yellowing and thickening of leaves forming **a net work of veins and veinlets in the infected leaves**
- Initially the leaves exhibit **only yellow coloured veins but under severe infection, the leaves become completely chlorotic and turn yellow**.

Yellow Vein Mosaic - *Yellow Vein Mosaic Virus* (YVMV)

Symptoms contd..

- The chlorophyll content of the leaves is reduced.
- Infected plants produce **very few small sized leaves, deformed and give a stunted look.**
- Affected plants produce dwarfed, malformed, **yellowish green or pale coloured fruits** and fetch less market price

Yellow Vein Mosaic - *Yellow Vein Mosaic Virus* (YVMV)

Virus

- The disease is caused by *Yellow Vein Mosaic Virus* (YVMV) and belongs to the **gemini virus.**
- The size of virus is 18 x 30 nm.
- Virus particles are spherical, isometric measuring 28-30 nm.

Transmission

- Virus is **not sap transmissible but under artificial conditions**, **it can be transmitted by grafting.**
- In nature, virus is transmitted by **white fly *Bemisia tabaci*.**

Yellow Vein Mosaic - *Yellow Vein Mosaic Virus* (YVMV)

Disease cycle and epidemiology:

- In nature, the virus is transmitted by insect vector, white fly (*Bemisia tabaci*) in a **persistent manner**.

- Both wild and cultivated plants serve as the source of inoculum and the incidence of the disease in a particular area depends upon the prevalence of wild or cultivated host plants, population buildup of the vectors and environmental conditions.

- Dry hot weather with little or no rainfall was conducive for disease development and also for the multiplication of the vector population

Yellow Vein Mosaic - *Yellow Vein Mosaic Virus* (YVMV)

Management:

- Various cultural practices like **destruction of wild hosts** from in and around the field.
- **Avoidance of mixed cropping** of pumpkin.
- **Adjustment of date of sowing** to avoid the period of maximum population of the whiteflies.
- **Regular removal of affected plants upto 55 days** help in reducing the incidence of the disease.
- Use of **yellow coloured polythene mulch** significantly delays the appearance of the disease.

Yellow Vein Mosaic - *Yellow Vein Mosaic Virus* (YVMV)

Management contd…

- Use **resistant cultivars** like **Punjab Padmini, Punjab-8, Parbhani Kranti and Hissar Unnat.**
- **Field sanitation should be given top priority** as many weeds are alternate hosts for the white fly vector.
- Border or **barrier crops like maize, sorghum and bajra** (5 to 6 rows)
- **Application of insecticides** immediately after seed germination

Management contd…

- **Yellow sticky traps** can also be used for monitoring

- **Four sprays of Metasystox** (demeton-S-methyl) at 15, 30, 45 and 60 days after sowing have been reported effective in controlling the insect vectors and keeping the disease under check.

- Two to three foliar sprays of dimethoate at 0.1% at 10 days interval are also effective

- Soil application of carbofuran at 1.5 kg a.i./ha in nursery bed and also during transplanting of the seedlings is effective.

- **Alchoholic extracts** Datura, Agave, Ginger lower the spread of the disease

Management contd…

- **Leaf extracts of** Bouganvillia, Prosopis, Neem oil (3%) and Neem seed kernal extract (5%) were highly effective.
- Release of Parasitoids like ***Encarsia bimaculata, Encarsia formosa***
- The greenhouse white fly, ***Trialeurodes vaporarium*** gave effective control of *Bemisia tabaci.*
- Biopesticide ***Beauveria bassiana*** (effective on nymphs and adults).